

EXPERIENCE

The Home Depot, Machine Learning Engineer II

Oct 2024 - Current

- Designed and implemented a scalable system architecture for a Databricks proof of concept, providing seamless version control for data scientists and merchants through an automated CI/CD pipeline that directly influenced strategic decisions by the EVP/CIO.
- Spearheaded the creation of the "NL2SQL" functionality of an internal Gen BI Multiagent Chatbot, allowing data partners to gain insights on real data using natural language.
- Developed and productionized foundational Dataform pipelines for two strategic ML partnerships, enabling the "Smart Fulfillment" model for order optimization and a "Consumer Sub-segmentation" model to power targeted advertising.
- Engineered and maintained an automated weekly updates process, slashing manual effort by over 97% (from 6 hours to 10 minutes per week) and saving approximately 303 company hours annually.
- Championed team-wide adoption and enablement of Dataform by delivering regular knowledge transfers and creating scalable resources, increasing team self-sufficiency and reducing recurring support.

The Home Depot, Software Engineer I

July 2023 - Oct 2024

- Led the metadata auditing workstream, scaling a proof of concept into a production-ready solution, now impacting multiple teams enterprise-wide.
- Enabled Dataform adoption across 150+ GCP projects, providing technical support and creating a template repository with built-in CI/CD, speeding up productionization for users.
- Created AI enabled automation scripts that reduced manual task times by up to 90%, cutting a repeatable process from 1 hour to 5 minutes.
- Contributed shared GAR components that get used in an ML Anomaly Detection process across all EDW data daily.
- Organized and hosted team-wide meetings, including the DNA Celebration and staff meetings, fostering collaboration and knowledge sharing across teams. Mentored new team members and introduced them to various technologies.

Computer Science Instructor, Nova Academy

Oct 2021 - Feb 2022

Nova Academy is an education network.

- Curated and taught a curriculum covering basic programming topics and fundamental engineering skills to ignite STEM interest in young developers. Topics covered include conditionals, variables, and functions.

PROJECTS

OpenTuner

Feb 2026 - Current

OpenTuner is an IOS Guitar Tuner and Metronome.

- Spearheaded the end-to-end development of "OpenTuner," an iOS application, achieving a 5-star rating on the App Store for its user-centric and ad-free experience.
- Engineered a custom real-time pitch detection algorithm using Swift and Core Audio, optimizing it for a diverse range of acoustic and electric instruments.
- Drove the full product lifecycle, from initial concept and UI/UX design in Figma to deployment and iterative updates based on user feedback.
- Enhanced the app's feature set by implementing a professional-grade metronome and a system for users to create and save custom instrument tunings.

EDUCATION

BS, Computer Science, Loyola Marymount University - Los Angeles, CA (2022)

- Cum Laude, LMU Arrupe Scholarship (2018-22), Dean's List, Phi Delta Theta

MS, Computer Science, Loyola Marymount University - Los Angeles, CA (2023)